

11

NUMBER AND ALGEBRA

SIMULTANEOUS EQUATIONS

OPTIONAL PATH TOPIC
RECOMMENDED FOR STAGE 6 MATHEMATICS ADVANCED

Many scientific, natural, economic and social phenomena can be modelled by equations. Often these models consist of more than one equation. For example, when manufacturing milk, equations can be written that describe relationships between quantity, cost and income. These equations can then be solved simultaneously to obtain information about pricing and the quantities that need to be produced and sold to make the most profit.

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Chapter outline

	Working mathematically				
11.01 Solving simultaneous equations graphically*	U	F		R	C
11.02 The elimination method*	U	F			
11.03 The substitution method*	U	F			
11.04 Problems involving simultaneous equations*	U	F	PS	R	C

*PATH

Wordbank

coefficient The numerical part of an algebraic term. For example, in $3x^2 + 7x - 1$ the coefficient of x is 7

elimination method A method of solving simultaneous equations that involves combining them to eliminate one of the variables

Quiz
Wordbank 11

graphical method A method of solving simultaneous equations that involves graphing them on a number plane and identifying the point(s) of intersection

simultaneous equations 2 (or more) equations that must be solved together so that the solution satisfies both equations. For example, $y = 2x + 1$ and $y = 3x$ are simultaneous equations that have a solution of $x = 1$, $y = 3$

substitution method A method of solving simultaneous equations that involves substituting one equation into another equation

U = Understanding | **F** = Fluency | **PS** = Problem solving | **R** = Reasoning | **C** = Communication

Videos (3):
11.02, 11.03 Simultaneous equations 1 • Simultaneous equations 2
11.04 Simultaneous equations problems

***Twig* video (1):**
11.04 The chase

***PhET* interactive (1):**
11.02 Equality explorer: Two variables

Quizzes (2):
- Wordbank 11
- SkillCheck 11

Worksheets (3):
11.01 Testing simultaneous equations • Intersection of lines
11.04 Simultaneous equations problems

Puzzles (2):
11.03 Simultaneous equations order activity • Simultaneous equations by substitution

Technology (1):
11.01 Simultaneous equations solver

Spreadsheet (1):
11.01 Simultaneous equations solver

Presentation (1):
11.04 Simultaneous equations

Nelson MindTap

To access resources above, visit **cengage.com.au/nelsonmindtap**

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In this chapter you will:

- (PATH) solve linear simultaneous equations graphically, including using graphing technology
- (PATH) solve linear simultaneous equations algebraically using the elimination and substitution methods
- (PATH) solve problems using linear simultaneous equations

Quiz
SkillCheck 11

SkillCheck ANSWERS ON P. 632

1 Given the equation $y = 2x + 5$, find y when:

a $x = 0$ **b** $x = 4$ **c** $x = \frac{1}{2}$ **d** $x = -3$

2 Given the equation $y = 4 - 3x$, find y when:

a $x = 5$ **b** $x = 1$ **c** $x = -1$ **d** $x = -\frac{1}{2}$

3 By completing a table of values, graph each equation.

a $y = x + 1$ **b** $y = 3x$ **c** $y = \frac{x}{2} - 1$

d $y = 3 - x$ **e** $x + y = 4$ **f** $2x - y = 5$

4 Test whether the point $(-2, 3)$ lies on the line represented by each equation.

a $y = 1 - x$ **b** $x + y = 3$ **c** $2x - y = 7$

d $\frac{1}{2}x + y = 2$ **e** $y = 3x + 7$ **f** $2y = 3x$

5 **a** Show that the point (2, 5) lies on both the lines $y = 2x + 1$ and $x + y = 7$.

b At what point do these 2 lines intersect?

6 Use the y-intercept and the gradient to graph each equation.

a $y = -2x + 3$ **b** $y = \frac{5}{2}x - 2$ **c** $y = -\frac{4}{3}x + 5$

7 Expand each expression.

a $3(2x + 4)$ **b** $-(x - 5)$ **c** $5(x + 2y - 3)$

8 Solve each equation.

a $6x = 2x + 8$ **b** $5x + 4 = 2x - 7$ **c** $2x + 3(x - 4) = 3$

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Solving simultaneous equations graphically

11.01

A linear equation such as $3x + 5 = 11$ is in one variable (x) and has only one solution ($x = 2$). However, a linear equation in **2** variables, such as $x + 3y = 5$, has more than one solution (for example, $x = 2$, $y = 1$, or $x = 5$, $y = 0$, and so on). The equation actually has an **infinite number** of solutions.

We will now look at solving 2 equations simultaneously to see if there is a solution that satisfies **both** equations. These are called **simultaneous equations**, which can be solved **graphically** or **algebraically**.

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Worksheets
Testing simultaneous equations

Intersection of lines

Technology
Solving simultaneous equations

Spreadsheet
Simultaneous equations solver

Solving simultaneous equations graphically

- Linear simultaneous equations can be graphed as lines on the same number plane.
- If 2 lines are drawn, the lines will intersect (unless they are parallel).
- At the point of intersection, the x-coordinate and y-coordinate represent the solution to the simultaneous equations.

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Example 1

On the same set of axes, graph $3x + y = 4$ and $x + y = -2$, then solve the equations simultaneously.

Solution

Step 1

Construct tables of values.

$3x + y = 4$

x	0	1	2
y	4	1	−2

$x + y = -2$

x	0	1	2
y	−2	−3	−4

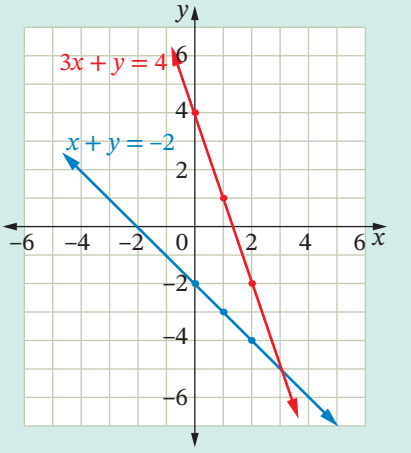
Step 2

Graph the equations.

The lines intersect at $(3, -5)$.

$\therefore$ The solution of the simultaneous equations $3x + y = 4$ and $x + y = -2$ is $x = 3, y = -5$.

Check that $x = 3, y = -5$ satisfies both equations.



EXERCISE 11.01 ANSWERS ON P. 634

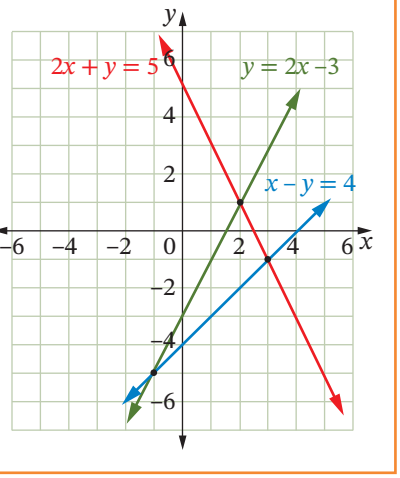
Solving simultaneous equations graphically U F R C

1 Use the graph to write the solution to each pair of simultaneous equations. C

a $x - y = 4$ and $2x + y = 5$

b $2x + y = 5$ and $y = 2x - 3$

c $x - y = 4$ and $y = 2x - 3$



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2 Graph each pair of equations on the same set of axes. Then find the solution to the pair of simultaneous equations.

a $y = 2x$ and $y = 3 - x$
b $y = 2x + 1$ and $y = x - 4$
c $x + y = 3$ and $4x + y = 6$
d $y = -x + 2$ and $y = 3x + 4$
e $y = 2x - 5$ and $y = 5x + 1$
f $2x + y = 6$ and $y = 1 - x$
g $y = 7 - x$ and $y = 3x + 5$
h $x + 2y = 7$ and $2x - y = 4$
i $3x - 2y = 12$ and $x + 2y = 8$
j $y = x + 3$ and $2x - y = 2$
k $5x - y = 5$ and $x + y = 4$
l $5x + 3y = 20$ and $y = x - 4$

3 **a** On the same set of axes, draw the graphs of $y = 1 - 2x$ and $2x + y = 4$.
b Why isn't there a solution to the simultaneous equations $y = 1 - 2x$ and $2x + y = 4$? R C

4 **a** On the same set of axes, draw the graphs of $x + y = 3$ and $2x + 2y = 6$.
b Why are there an infinite number of solutions to the simultaneous equations $x + y = 3$ and $2x + 2y = 6$? R C

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EXAMPLE 1

The elimination method 11.02

Using graphs to solve simultaneous equations can be time-consuming and inaccurate.

Algebraic methods provide a better way of solving them.

There are 2 algebraic methods: the **elimination method** and the **substitution method**.

In the elimination method, equations are added or subtracted to eliminate one of the variables.

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Interactive
Equality explorer: Two variables

Video
Simultaneous equations 1

Example 2

Solve the simultaneous equations $x + 3y = 7$ and $4x - 3y = 13$ using the elimination method.

Solution

Label each equation.

$x + 3y = 7$ [1]

$4x - 3y = 13$ [2]

Since there is the same number of ys in each equation, and since they are **opposite in sign** ($3y$ and $-3y$), add equations [1] and [2] to eliminate the variable y.

$5x = 20$ [1] + [2]

$\frac{5x}{5} = \frac{20}{5}$

$\therefore x = 4$

Substitute $x = 4$ into equation [1] to find the y-value.

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$$x + 3y = 7$$
$$4 + 3y = 7$$
$$4 + 3y - 4 = 7 - 4$$
$$3y = 3$$
$$\therefore y = 1$$

$\therefore$ The solution is $x = 4, y = 1$.

Check that this solution works for both equations [1] and [2].

Example 3

Solve the simultaneous equations $2k + 3m = 9$ and $2k - 5m = 1$.

Solution

Label each equation.

$2k + 3m = 9$ [1]

$2k - 5m = 1$ [2]

Since there is the same number of ks in each equation, and because **they have the same sign** ($2k$ and $2k$), subtract equation [2] from [1] to eliminate k.

$8m = 8$ [1] – [2]

$$\frac{8m}{8} = \frac{8}{8}$$

$\therefore m = 1$

Substitute $m = 1$ into equation [1] to find the value of k.

$$2k + 3m = 9$$
$$2k + 3 \times 1 = 9$$
$$2k + 3 = 9$$
$$2k + 3 - 3 = 9 - 3$$
$$2k = 6$$
$$\frac{2k}{2} = \frac{6}{2}$$
$$\therefore k = 3$$

$\therefore$ The solution is $m = 1, k = 3$.

Check that this solution works for both equations [1] and [2].

Example 4

Solve $3a + 4c = 8$ and $2a - 3c = 11$.

Solution

Label each equation.

$3a + 4c = 8$ [1]

$2a - 3c = 11$ [2]

In this case, neither adding nor subtracting equations [1] and [2] will eliminate a variable. Let's choose to eliminate c. We need to make the **coefficient** of c the same in both equations ($12c$).

The **coefficient** of c is the number in front of the c in the equation

$9a + 12c = 24$ [3] Multiplying both sides of [1] by 3

$8a - 12c = 44$ [4] Multiplying both sides of [2] by 4

$17a = 68$ [3] + [4]

$\therefore a = 4$

Substitute $a = 4$ in [1] to find c.

$3a + 4c = 8$

$3 \times 4 + 4c = 8$

$12 + 4c = 8$

$4c = -4$

$c = -1$

$\therefore$ The solution is $a = 4$, $c = -1$.

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Video
Simultaneous equations 2

11.02

EXERCISE 11.02 ANSWERS ON P. 634

The elimination method U F

EXAMPLE 2

1 For each pair of simultaneous equations, eliminate one variable by adding the equations, then solve the equations.

a $4k + d = 5$
$2k - d = 7$

b $2x - w = 6$
$x + w = 9$

c $3g + 5h = 4$
$2g - 5h = 6$

d $7p - 4n = -20$
$3p + 4n = 10$

e $4q + 3r = 8$
$-q - 3r = 7$

f $-5k - 3x = 8$
$5k + 4x = -3$

g $-4c - 6e = -12$
$4c - 10e = -4$

h $-3y + 5k = 21$
$3y + k = -3$

i $a + 3f = 8$
$-a + 4f = 6$

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EXAMPLE 3

2 For each pair of simultaneous equations, eliminate one variable by subtracting the equations, then solve the equations.

a	$5k + d = 16$ $3k + d = 4$	**b**	$4a + 3c = 7$ $a + 3c = 4$	**c**	$4h + 3y = 24$ $4h - y = 8$
d	$3x + 5e = 16$ $3x - 2e = -5$	**e**	$4q - 2w = -1$ $7q - 2w = 8$	**f**	$6p + 5c = 39$ $4p + 5c = 31$
g	$5y + 3m = 18$ $2y + 3m = 6$	**h**	$3a + 2r = 8$ $a + 2r = 10$	**i**	$-x + 5w = 8$ $-x + 3w = 4$
j	$4y + 7g = 2$ $4y - 3g = 22$	**k**	$2e - 3n = 14$ $5e - 3n = -1$	**l**	$7k - 5h = 31$ $7k + 3h = 43$

EXAMPLE 4

3 Solve each pair of simultaneous equations.

a	$3w + q = 6$ $2w - 3q = 15$	**b**	$2x + m = 5$ $3x + 2m = 3$	**c**	$2d + 3h = 25$ $d + 4h = -5$
d	$-3g + 2n = 9$ $g + 5n = 14$	**e**	$5m - h = 10$ $m - 3h = 2$	**f**	$2y + 3e = -6$ $5y - 2e = 23$
g	$3q - 2w = 11$ $2q - 5w = 22$	**h**	$5a + 3d = 4$ $4a + 2d = 3$	**i**	$-2p + 3k = 19$ $7p + 4k = 6$
j	$5a + 2f = -14$ $2a - 3f = 2$	**k**	$5r - 3c = 2$ $-3r + 2c = -14$	**l**	$5y - 4x = 1$ $2y - 3x = 6$
m	$3x + 4y = 20$ $2x - 5y = 21$	**n**	$7g + 3h = 39$ $3g + 5h = 26$	**o**	$5w - 3k = 25$ $3w - 7k = 28$

11.03 The substitution method

PATH

Video
Simultaneous equations 1

Puzzles
Simultaneous equations order activity

Simultaneous equations by substitution

With the **substitution method**, **substitute** the x or y variables from one equation into the other equation.

Example 5

Solve the simultaneous equations $y = x + 4$ and $y = 3x - 2$.

Solution

Label each equation.

$y = x + 4$ [1]

$y = 3x - 2$ [2]

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Use equation [1] to **substitute for *y*** in equation [2] and solve for x.

$$x + 4 = 3x - 2$$
$$x + 4 - 3x = 3x - 2 - 3x$$
$$-2x + 4 = -2$$
$$-2x + 4 - 4 = -2 - 4$$
$$-2x = -6$$
$$\frac{-2x}{-2} = \frac{-6}{-2}$$
$$x = 3$$

Now substitute $x = 3$ into equation [1] to find y.

$$y = x + 4$$
$$y = 3 + 4$$
$$= 7$$

$\therefore$ The solution is $x = 3$ and $y = 7$.

Example 6

Solve the simultaneous equations $5x + 3y = 9$ and $y = 7 - 3x$.

Solution

Label each equation.

$5x + 3y = 9$ [1]

$y = 7 - 3x$ [2]

Since y is the subject in [2], **substitute equation [2]** into equation [1] to give an equation using x only.

$$5x + 3(7 - 3x) = 9$$
$$5x + 21 - 9x = 9$$
$$-4x = -12$$
$$\frac{-4x}{-4} = \frac{-12}{-4}$$
$$x = 3$$

Now substitute $x = 3$ into equation [2] to find y.

$$y = 7 - 3x$$
$$y = 7 - 3 \times 3$$
$$= -2$$

$\therefore$ The solution is $x = 3$ and $y = -2$.

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11.03

Video
Simultaneous equations 2

PATH

EXERCISE 11.03 ANSWERS ON P. 635

The substitution method U F

EXAMPLE 6

1 Use the substitution method to solve each pair of simultaneous equations.

a $y = 2x + 1$ and $y = x + 3$

b $y = 5 - 2x$ and $y = 3x + 2$

c $x = 3 + 2y$ and $x = 9 - y$

d $y = -x$ and $y = 3x - 8$

e $x = 1 - 4y$ and $x = 2y + 7$

f $x = 2y$ and $x = 6 - y$

EXAMPLE 6

2 Solve each pair of simultaneous equations.

a $y = 2x + 3$ and $3x - y = 6$

b $y = x - 2$ and $3x + y = 18$

c $y = 1 - 4x$ and $4x + 2y = 3$

d $x = 2y - 5$ and $4x - y = -13$

e $x = 3y - 4$ and $5x - 4y = 2$

f $x = 5 - 3y$ and $4y - x = 23$

g $2x - 5y = -1$ and $y = 10 - x$

h $6y - 2x = 9$ and $y = \frac{x+2}{2}$

i $x = \frac{9-y}{3}$ and $3x + 2y = 10$

j $y = 3x + 5$ and $4x - 3y = 1$

11.04 Problems involving simultaneous equations

PATH

Video
The chase

Worksheet
Simultaneous equations problems

Presentation
Simultaneous equations

Sometimes, worded problems can be solved using simultaneous equations. In these situations, following these steps:

- Read the problem carefully
- Identify the variables to be used
- Use the variables to write simultaneous equations from the information given in the problem
- Solve the equations
- Solve the problem by answering in words

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Example 7

At an art show there were 520 guests. If there were 46 more women than men, how many women attended the show?

Solution

Let the number of women attending be w.

Let the number of men attending be m.

$w + m = 520$ [1] 520 people altogether.

$w = m + 46$ [2] 46 more women than men.

Use equation [2] to substitute for w in equation [1].

$m + 46 + m = 520$

$2m + 46 = 520$

$2m + 46 - 46 = 520 - 46$

$2m = 474$

$m = 237$

Substitute $m = 237$ into equation [2] to find w.

$w = 237 + 46$

$= 283$

$\therefore$ There were 283 women who attended the art show.



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Video
Simultaneous equations problems

Example 8

Masi and Amir spent $1582 on shrubs and trees to plant on their large block of land. Altogether they bought 73 plants. The shrubs cost $19 each while the trees cost $32 each. How many of each plant did they buy?

Solution

Let x be the number of shrubs.

Let y be the number of trees.

$\therefore x + y = 73$ [1]

and $19x + 32y = 1582$ [2]

Neither adding nor subtracting equations [1] and [2] will eliminate a variable.
Let's choose to eliminate x.

We will need to make the coefficient of x the same in both equations ($19x$).

$19x + 32y = 1582$ [2]

$19x + 19y = 1387$ [3] Multiplying both sides of equation [1] by 19.

$13y = 195$ [2] – [3]

$y = 15$

Substitute $y = 15$ in [1] to find the value of x.

$x + y = 73$

$x + 15 = 73$

$x = 58$

So Masi and Amir bought 58 shrubs and 15 trees.

EXERCISE 11.04 ANSWERS ON P. 635

Problems involving simultaneous equations U F PS R C

EXAMPLE 7

1 At a school concert there were 640 guests. There were 70 more women than men. How many of the audience were men? R C

2 At a circus, there were twice as many children as there were adults in attendance. Altogether, 1020 attended the circus. How many were children? R C

EXAMPLE 8

3 Tickets to a concert cost $15 for children and $28 for adults. Altogether, 650 people attended the concert and ticket sales totalled $13 052. Let a stand for the number of adults and c stand for the number of children in the audience. R C

a Explain why the equations $a + c = 650$ and $28a + 15c = 13\,052$ correctly match the information.

b Solve the equations simultaneously to find the number of children that attended the concert.

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11.04

4 Ashleigh bought a total of 130 meat pies and sausage rolls for the canteen of the local football club. Each meat pie cost her $3 and the sausage rolls were $2 each. Altogether, Ashleigh spent $335. How many sausage rolls did she buy? PS R C

5 Tayyab is 3 times as old as Sejuti. The sum of their ages is 48. How old are Tayyab and Sejuti? PS R C

6 The sum of the ages of Hayley and her mother is 70. The difference between their ages is 38 years. How old is Hayley? PS R C

7 A business bought a total of 60 ink cartridges. Some of them were black, costing $35 each. The others were colour, each costing $49. How many of each type did the business buy if the total cost of the ink cartridges was $2422? PS R C

8 The cost of going to the movies for 3 adults and 5 children is $142, while the cost for 5 adults and 8 children is $231.50. Find the cost for an adult and the cost for a child. PS R C

9 Pete's Pizzas sells Supreme pizzas for $15.90 each and vegetarian pizzas for $13.50 each. If 45 pizzas were sold at lunchtime, totalling $684.30, how many of each pizza were sold? PS R C

10 Ronel bought 4 punnets of strawberries and 7 punnets of blueberries for $40.30 and Nikita bought 5 punnets of strawberries and 2 punnets of blueberries for $26.75. What was the cost of each punnet of strawberries and blueberries? PS R C

11 A money box contains only 20-cent coins and 50-cent coins. Altogether, there are 853 coins in the money box and they amount to $281. Let x be the number of 20c coins and y be the number of $2 coins. R C

a Explain why the equations $x + y = 853$ and $20x + 50y = 28\,100$ correctly match the information.

b Solve the equations to determine the number of 20-cent and 50-cent coins in the money box.

12 The initial cost for producing bottles of fresh orange juice is $135 plus $1.20 for each bottle. The bottles of juice are sold for $3 each. C is the cost in dollars, R is the total sales in dollars and n is the number of bottles produced and/or sold. PS R C

a Explain why the equations $C = 135 + 1.2n$ and $R = 3n$ correctly match the information.

b Copy and complete the tables of values below for both equations.

$C = 135 + 1.2n$

n	0	50	100
C			

$R = 3n$

n	0	50	100
R			

c Draw the graphs of both equations on the same axes for values of 0 to 100 for n on the horizontal axis and values of $0 to $300 on the vertical axis.

d For what value of n is total sales equal to total cost (the break-even point)?

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